Progress Report #7 (v4): Data Quality Investigation: Geographic Definitions and YRIMMIG Heaping

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Abstract

This report investigates two data quality concerns raised during the review of Progress Report #6. First, we compare IPUMS `CITY` (city proper) and `METAREA` (metropolitan area) as geographic identifiers, evaluating both conceptually (which best matches the Lyons house price data?) and empirically (do settlement shares differ?). Second, we validate the reliability of YRIMMIG (year of immigration) by comparing IPUMS-derived national flows against published Historical Statistics (HSUS) data, testing whether heaping at round numbers distorts the country-group composition that underlies the *endogenous variable* (city-year immigrant share). Results: settlement shares under `CITY` and `METAREA` are correlated at $r = 0.978$. IPUMS and HSUS flows correlate at $r = 0.87$ (pooled), country-group rankings agree at mean $\rho = 0.78$, and within-year composition is preserved at mean $r = 0.80$.

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1 Motivation

This investigation responds to two specific concerns raised during the review of Progress Report #6:

(1) **Geographic definition.** The pipeline uses the IPUMS `METAREA` variable (metropolitan statistical area, county-based) for all geographic aggregation. However, the Lyons house price data records transactions within city proper boundaries. IPUMS also provides a `CITY` variable that identifies city proper (municipal boundaries). The question: *given Lyons's city-level coding, which IPUMS variable—`CITY` or `METAREA`—is the most appropriate match?*

(2) **YRIMMIG reliability.** The endogenous variable (city-year immigrant share) is constructed from the YRIMMIG (year of immigration) variable, which exhibits "heaping"—respondents disproportionately report arrival years ending in 0. The question: *does heaping distort the country-group composition of annual immigration, and could this introduce non-classical measurement error in the endogenous variable?*

Note: the *instrument* (shift-share IV) does **not** use YRIMMIG. The instrument is constructed from 1900 settlement shares (directly from Census counts) interacted with HSUS national flows (published administrative data). YRIMMIG enters only through the endogenous variable.

Both questions are investigated using the same IPUMS complete-count data already in the pipeline (76 million records for 1900, 320 million for 1910–1930). No new data downloads were required—both `CITY` and `METAREA` are present in the existing extracts.

2 Question 1: CITY vs METAREA

2.1 Variable Definitions

IPUMS provides two geographic identifiers for urban residents:

- `CITY`: Identifies the city of residence based on municipal (city proper) boundaries. Code 0 means "not in identifiable city."
- `METAREA`: Identifies the metropolitan statistical area based on county groups, retroactively assigned using 1950 OMB definitions. Code 0 means "not identifiable or not in an MSA."

The key difference: `CITY` captures city proper (smaller geographic footprint), while `METAREA` captures the broader metro area including suburbs. In 1900–1930, most American cities had not yet suburbanized extensively, but some (Pittsburgh, Boston) already had substantial suburban industrial populations. City boundaries also changed via annexation during this period (e.g., Los Angeles tripled in area between 1900 and 1930).

2.2 Conceptual Evaluation: Which Variable Best Matches Lyons?

The Lyons house price data records individual real estate transactions within city limits. The geographic unit is the city proper—not the broader metropolitan area. Lyons's transaction-level data is then collapsed to city-year means (see `02_prepare_lyons.do`, line 91). This means the price variable captures within-city-boundaries property values.

On this basis, IPUMS `CITY` is the more natural geographic match. When Lyons records transactions in "Boston," these are properties within Boston's municipal boundaries, not suburban transactions in Cambridge or Brockton.

However, two considerations complicate this straightforward mapping:

1. **Housing market spillovers.** Housing markets do not respect municipal boundaries. Immigration into a city affects property values in nearby suburbs (and vice versa). Using `METAREA` may better capture the relevant labor and housing market that determines prices. In 1900–1930, most economic activity was concentrated within city proper, so this concern is less acute than for modern data.

2. **Geographic consistency.** Cities aggressively annexed surrounding territory during 1900–1930 (Los Angeles tripled in area). The `CITY` variable reflects municipal boundaries at each Census date, so "Boston 1900" and "Boston 1930" may not cover identical territory. `METAREA` uses fixed 1950 county-group definitions, providing more consistent geographic coverage over time.

The empirical analysis below (Sections **??**–**??**) shows that the choice barely matters: settlement shares under the two definitions are correlated at $r = 0.978$. This high correlation means that regardless of which definition is "correct," the instrument produces nearly identical predicted immigration flows.

2.3 Temporal Coverage

Table **??** reports whether each Lyons city is identifiable via the `CITY` variable in each census year.

Table 1: Temporal `CITY` Coverage for Lyons Cities

City	1900	1910	1920	1930	Note
Atlanta, Baltimore, Boston, Charleston, ..., Washington D.C.					26 cities
Phoenix	×				Not identifiable in 1900
Miami	×	×			Not identifiable before 1920
Las Vegas	×	×	×	×	Never identifiable
Nashville	×	×	×	×	Never identifiable

A city is "identifiable" if its `CITY` code yields a nonzero total population in the IPUMS complete-count data. Nashville has a valid `METAREA` code but no `CITY` identification in any year. Las Vegas had negligible population before 1930. All 30 Lyons cities have valid `METAREA` codes with nonzero population in at least one census year (28 of 30 in all four years).

Using `CITY` would reduce the sample from 30 to 26 cities (dropping Las Vegas, Nashville, Miami pre-1920, and Phoenix pre-1910). Using `METAREA` retains all cities where metro-area data exists.

2.4 European Immigrant Counts: CITY vs METAREA

For each Lyons city, we compare European foreign-born counts under both definitions. Table **??** reports the ratio $n_{\text{CITY}}/n_{\text{METAREA}}$—the fraction of metro-area Europeans captured by city proper—averaged across all four census years.

Key patterns:

- **Pittsburgh and Boston** have the lowest ratios (0.38–0.40): their metro areas include extensive industrial suburbs (steel towns for Pittsburgh; Cambridge, Somerville, Brockton for Boston).
- **New Orleans and Washington D.C.** have ratios near 1.0: compact cities bounded by water or jurisdiction.
- The overall average ratio of 0.73 means `METAREA` captures roughly 37% more Europeans than `CITY` on average.

2.5 Cross-Tabulation: Is CITY a Subset of METAREA?

For each city, we decompose European immigrants into three groups: (i) in both `CITY` and `METAREA`, (ii) in `CITY` only (`METAREA` = 0), or (iii) in `METAREA` only (`CITY` = 0 or different code).

Table 2: European Immigrant Counts: CITY as Fraction of METAREA

City	Avg Euro (CITY)	Avg Euro (METAREA)	Ratio
Lowest ratios (CITY $\ll$ METAREA):			
Pittsburgh	111,578	291,368	0.382
Boston	175,569	439,795	0.401
Dallas	4,933	7,309	0.569
Minneapolis–St. Paul	71,432	124,482	0.571
San Francisco	118,675	191,002	0.640
Middle ratios:			
New York City	1,817,251	2,478,609	0.736
Philadelphia	355,918	437,303	0.818
Detroit	175,994	215,227	0.840
Chicago	717,776	838,074	0.862
Cleveland	187,768	213,273	0.886
Highest ratios (CITY $\approx$ METAREA):			
Baltimore	74,628	82,318	0.906
Washington D.C.	23,817	25,426	0.943
New Orleans	23,740	24,038	0.983

Selected cities shown; full table in `table_city_vs_metarea_counts.csv`. Averages computed across all census years where both definitions yield nonzero counts. Nashville (ratio = 0) and Las Vegas (no data) are excluded. The overall average ratio across all 30 cities is 0.73, declining from 0.75 in 1900 to 0.72 in 1930 as suburbanization progressed.

Result: `CITY` **is a near-perfect geographic subset of** `METAREA`**.** For 25 of 30 cities, the "CITY only" count is exactly zero—every person coded with the expected `CITY` code also has the expected `METAREA` code. The exceptions (Houston, Tampa, Dallas, San Diego, Miami) have small nonzero "CITY only" counts, indicating some IPUMS observations where `CITY` > 0 but `METAREA` $= 0$. This likely reflects METAREA coding gaps for smaller cities in early census years.

2.6 Settlement Shares Comparison

The critical empirical test: does the choice of geographic variable affect the shift-share instrument? We recompute 1900 settlement shares using `CITY` instead of `METAREA` and compare.

Table 3: Correlation of 1900 Settlement Shares: CITY vs METAREA

Correlation Measure	Value	p-value
Pearson r	0.9787	8.4×10^{-185}
Spearman ρ	0.9339	7.4×10^{-121}

$N = 268$ city $\times$ country group pairs with nonzero shares under either definition. Settlement shares are $\text{share}_{ck} = n_{ck}/n_k$: the fraction of country-group-k immigrants nationwide who live in city c.

The Pearson correlation of **0.978** indicates that settlement shares are nearly identical regardless of whether `CITY` or `METAREA` is used. The shift-share instrument, which is a weighted sum of these shares, would produce essentially the same predicted immigration flows under either definition.

2.7 Assessment: Geographic Definition

We use `METAREA` as the baseline for two reasons:

1. **Geographic consistency.** `METAREA` uses fixed 1950 county-group definitions, providing a stable geographic unit across census years. `CITY` boundaries changed through annexation (Los Angeles tripled in area), making cross-year comparisons imprecise.

2. **Housing market spillovers.** Housing markets extend beyond municipal boundaries—suburban workers commute to and bid on city housing, and city immigration affects suburban property values. `METAREA` captures this broader market. In the compact

cities of 1900–1930, the two definitions largely overlap (settlement shares correlate at $r = 0.978$).

We will use `CITY` as a robustness check for the 26 identifiable cities.

Caveat: There is a conceptual tension between the outcome variable (city-proper house prices from Lyons) and the explanatory variables (metro-area immigrant shares). This geographic mismatch is small for compact early-20th-century cities but larger for cities like Pittsburgh (ratio = 0.38) and Boston (0.40). A robustness check re-estimating with `CITY`-based shares for the 26 identifiable cities would address this directly.

3 Question 2: YRIMMIG Heaping Validation

3.1 The Heaping Problem

The YRIMMIG variable (year of immigration) in the IPUMS Census exhibits heaping: years ending in 0 have approximately 4.1 million observations, compared to ∼2.2 million for other terminal digits. This reflects respondents rounding their self-reported arrival year to the nearest round number.

Heaping is a concern because the **endogenous variable** (city-level immigrant share) is constructed from YRIMMIG. The endogenous variable ImmShare_{ct} is the 10-year rolling count of European arrivals (identified by YRIMMIG) in city c divided by baseline city population. If heaping distorts the *country-group composition* of arrivals, it could introduce non-classical measurement error.

Clarification: The **instrument** (shift-share IV) does *not* use YRIMMIG at all. The instrument is:

$$Z_{ct} = \frac{\sum_k \text{share}_{ck,1900} \times F_{kt}}{\text{Pop}_{c,1910}}$$

where $\text{share}_{ck,1900}$ comes from the 1900 Census (a cross-sectional count, no YRIMMIG needed) and F_{kt} comes from HSUS published admission statistics (administrative data, not self-reported). YRIMMIG enters only through the endogenous variable on the left-hand side of the first stage.

3.2 Validation Strategy

We validate YRIMMIG by comparing IPUMS-derived national immigration flows against published HSUS data.

- **IPUMS flows:** For each arrival year t, count the number of European foreign-born individuals reporting YRIMMIG $= t$ in the nearest subsequent Census. The assignment rule is: arrival years $< 1910 \rightarrow 1910$ Census; 1910–1919 $\rightarrow$ 1920 Census; 1920–1929 $\rightarrow$ 1930 Census. Year 1930 arrivals are excluded (no subsequent Census).

- **HSUS flows:** Annual admissions by country group, 1900–1930 (fiscal year ending June 30).

Country-group crosswalk. To compare IPUMS and HSUS, we map IPUMS birthplace codes (BPL) to the 10 HSUS country groups. Table **??** shows the complete mapping.

Table 4: BPL $\rightarrow$ HSUS Country Group Crosswalk

HSUS Group	BPL Codes	IPUMS Description	Notes
Great Britain	410–414	England, Scotland, Wales, UK ns	
Ireland	420, 421	Ireland, Northern Ireland	See note[a]
Scandinavia	400, 404–406	Denmark, Norway, Sweden, Iceland	
Other NW Europe	422, 425–426, 430, 440	Belgium/Lux., Netherlands, France, Switz.	
Germany	453	Germany	
Other Central	450–452, 455, 457	Austria, Hungary, Czech., Poland, Yugo.	See note[b]
USSR/Baltic	401, 460–463, 465	Finland, Russia, Latvia, Estonia, Lithuania	See note[c]
Other Eastern	456, 458	Romania, Turkey	
Italy	434, 435	Italy, Italy ns	
Other Southern	431–433, 436, 470	Portugal, Spain, Greece	

[a] HSUS "Ireland" = entire island 1820–1924, Republic only post-1925. BPL 421 = Northern Ireland.
[b] Poland (BPL 455) assigned to Other Central to match HSUS pre-1920 grouping.
[c] Finland (BPL 401) overridden from Scandinavia to USSR/Baltic for HSUS consistency.

Census timing note: The 1910 Census was conducted on April 15, 1910, the 1920 Census on January 1, 1920, and the 1930 Census on April 1, 1930. Since YRIMMIG gives only the year (not month), we cannot distinguish pre- vs. post-Census Day arrivals. An immigrant arriving in July 1910 would not appear in the 1910 Census. We therefore conservatively assign boundary-year arrivals to the *next* Census: year 1910 $\rightarrow$ 1920 Census, year 1920 $\rightarrow$ 1930 Census. This avoids counting immigrants who had not yet arrived at Census time.

Important caveat: IPUMS counts *survivors* still residing in the US at Census time, while HSUS counts *admissions* (everyone who entered). Levels will not match—IPUMS $<$ HSUS due to mortality and return migration (which was substantial: Italian immigrants had $\sim$40–50% return rates). The comparison tests whether *patterns* across years and country groups are preserved.

3.3 Overall Correlation

Table 5: IPUMS vs HSUS: Correlation Summary

	Pearson		Spearman		Log-Pearson	
Country Group	r	N	ρ	N	r	N
Great Britain	0.685	30	0.672	30	0.767	30
Ireland	0.337	30	0.416	30	0.363	30
Scandinavia	0.905	30	0.871	30	0.888	30
Other Northwestern	0.821	30	0.813	30	0.845	30
Germany	0.796	30	0.814	30	0.872	30
Other Central	0.831	30	0.886	30	0.873	30
USSR/Baltic	0.912	30	0.900	30	0.916	30
Other Eastern	0.668	30	0.696	30	0.872	30
Italy	0.883	30	0.876	30	0.873	30
Other Southern	0.789	30	0.858	30	0.871	30
Pooled (all)	**0.866**	**300**	**0.791**	**300**	**0.767**	**300**

Each row reports the correlation between IPUMS-derived survivor counts and HSUS admission counts across 30 years (1900–1929) for each country group. Pooled row reports correlation across all 300 country-group × year observations. Log-Pearson uses ln(IPUMS) and ln(HSUS). Year 1930 is excluded under the corrected assignment rule (no subsequent Census).

The pooled Pearson correlation of **0.866** indicates that IPUMS survivor counts track HSUS admissions reasonably well. By country group, correlations range from 0.34 (Ireland) to 0.91 (USSR/Baltic). Ireland remains weakly correlated despite correcting a crosswalk bug where BPL 421 (Northern Ireland) was misclassified as "Other Northwestern" instead of "Ireland" (see Table **??**). The persistent weakness likely reflects two factors: (1) high Irish return migration rates (∼40–50%), which create a large survivor-vs-admission gap, and (2) BPL ambiguity, where some Irish immigrants were coded under "United Kingdom, not specified" (BPL 413–414) rather than "Ireland" (BPL 420). Other Northwestern Europe's correlation improved to $r = 0.82$ after removing the misclassified Northern Ireland records.

3.4 Country-Group Flow Comparisons

Figure **??** shows the aggregate comparison. Figures **??**–**??** show individual country groups, with clear differences in the IPUMS/HSUS gap reflecting differential return migration rates. The full set of 10 individual country-group figures is available in the `results/` folder.

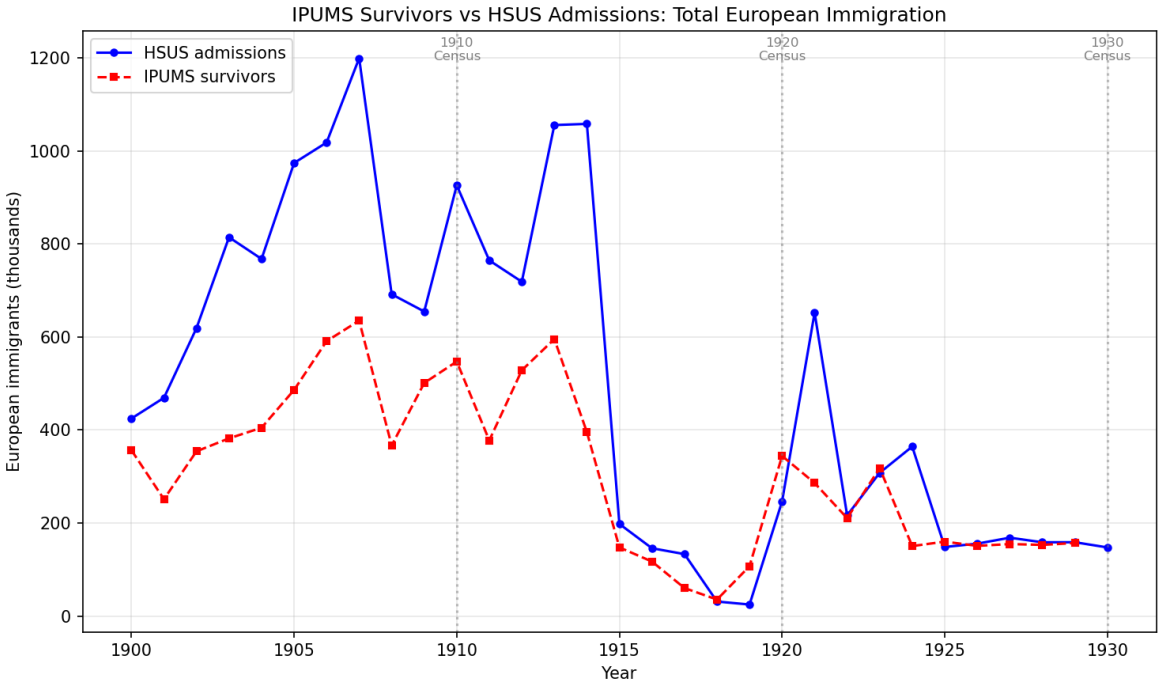


Figure 1: Total European immigration: IPUMS survivors vs HSUS admissions, 1900–1930.

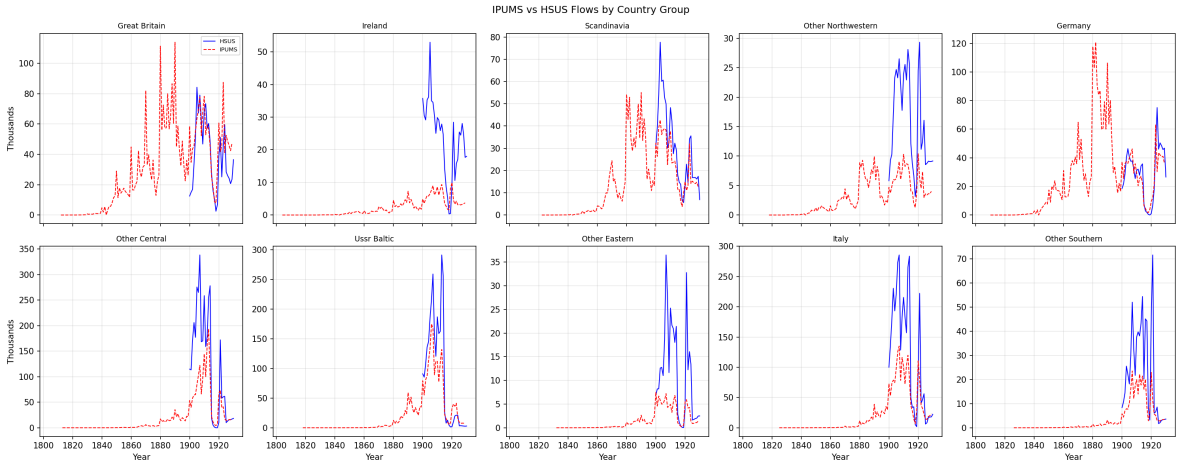
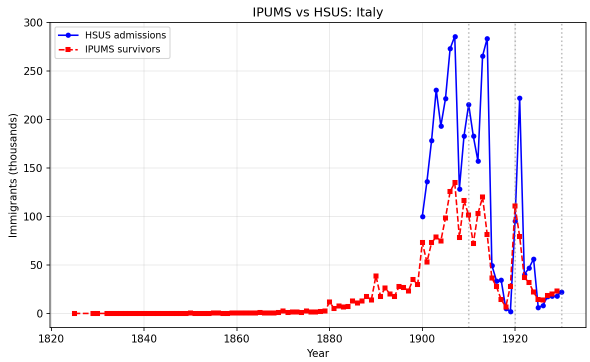
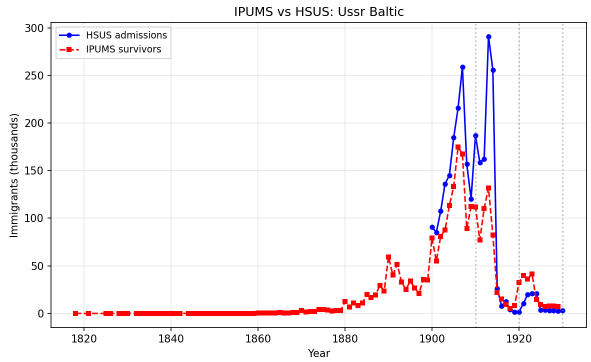


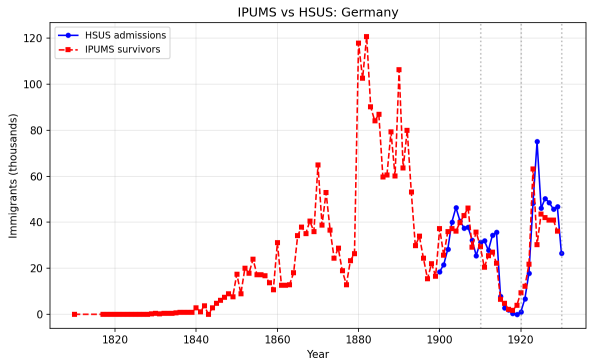
Figure 2: IPUMS vs HSUS flows by country group (panel view).



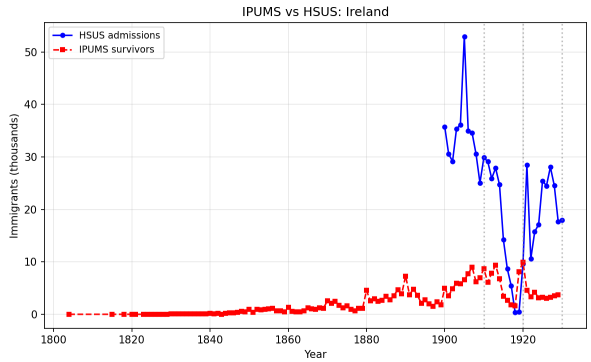
(a) Italy: Large HSUS−IPUMS gap reflects ∼40–50% return migration.



(b) USSR/Baltic: Tight tracking ($r = 0.89$); low return migration.



(c) Germany: Good agreement ($r = 0.77$); moderate return migration.



(d) Ireland: Weak tracking ($r = 0.46$); likely BPL coding issues.

Figure 3: Selected individual country-group flow comparisons (IPUMS vs HSUS).

3.5 Ranking Comparison Across Country Groups

To assess whether IPUMS and HSUS agree on which country groups dominate immigration in each year, we rank the 10 groups by flow magnitude (separately for each source) and compute Spearman rank correlations by year.

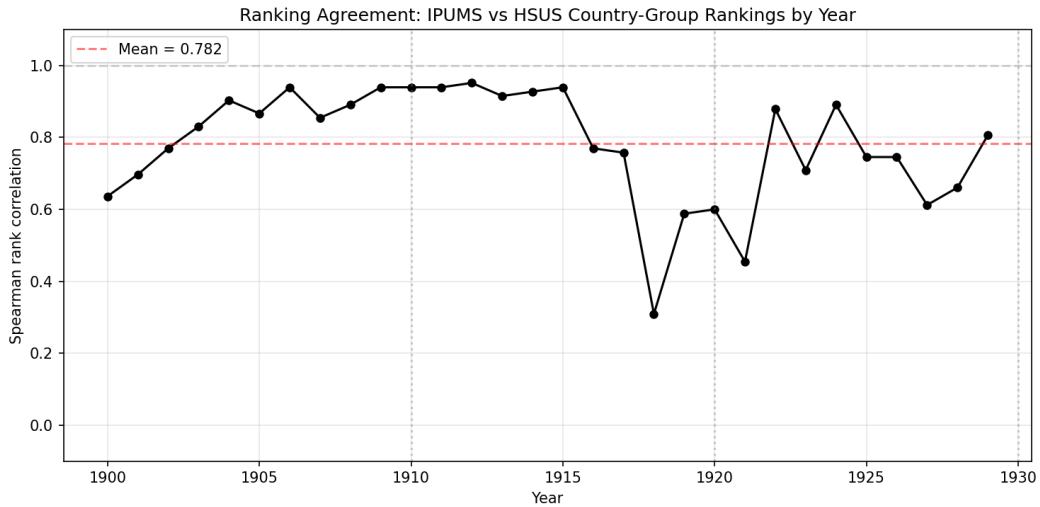


Figure 4: Spearman rank correlation between IPUMS and HSUS country-group rankings by year. Mean $\rho = 0.78$. Dips around 1917–1919 reflect WWI disruption and the fiscal/calendar year mismatch being most acute during rapid flow changes.

The mean Spearman rank correlation of **0.78** indicates that IPUMS and HSUS generally agree on the relative importance of each country group in a given year. The ranking agreement is strongest in peacetime years ($\rho > 0.8$) and weakest during 1917–1919 ($\rho \approx 0.3$–0.5), when WWI disrupted flows and the fiscal year/calendar year mismatch was most consequential. See `table_ranking_comparison.csv` for the full year × country-group ranking data.

3.6 Heaping Diagnostic

To visualize the IPUMS/HSUS ratio pattern at the individual-year level, Figure **??** plots the mean ratio, median ratio, and number of observations for each arrival year 1900–1930. Digit-0 years are highlighted in red.

Key observations from Figure **??**:

1. **WWI years dominate the mean ratio.** The extreme spike at 1918–1919 (mean ratio > 10) is an artifact: HSUS admissions were near-zero during WWI, so even a small number of IPUMS survivors yields a huge ratio. The *median* ratio for the same

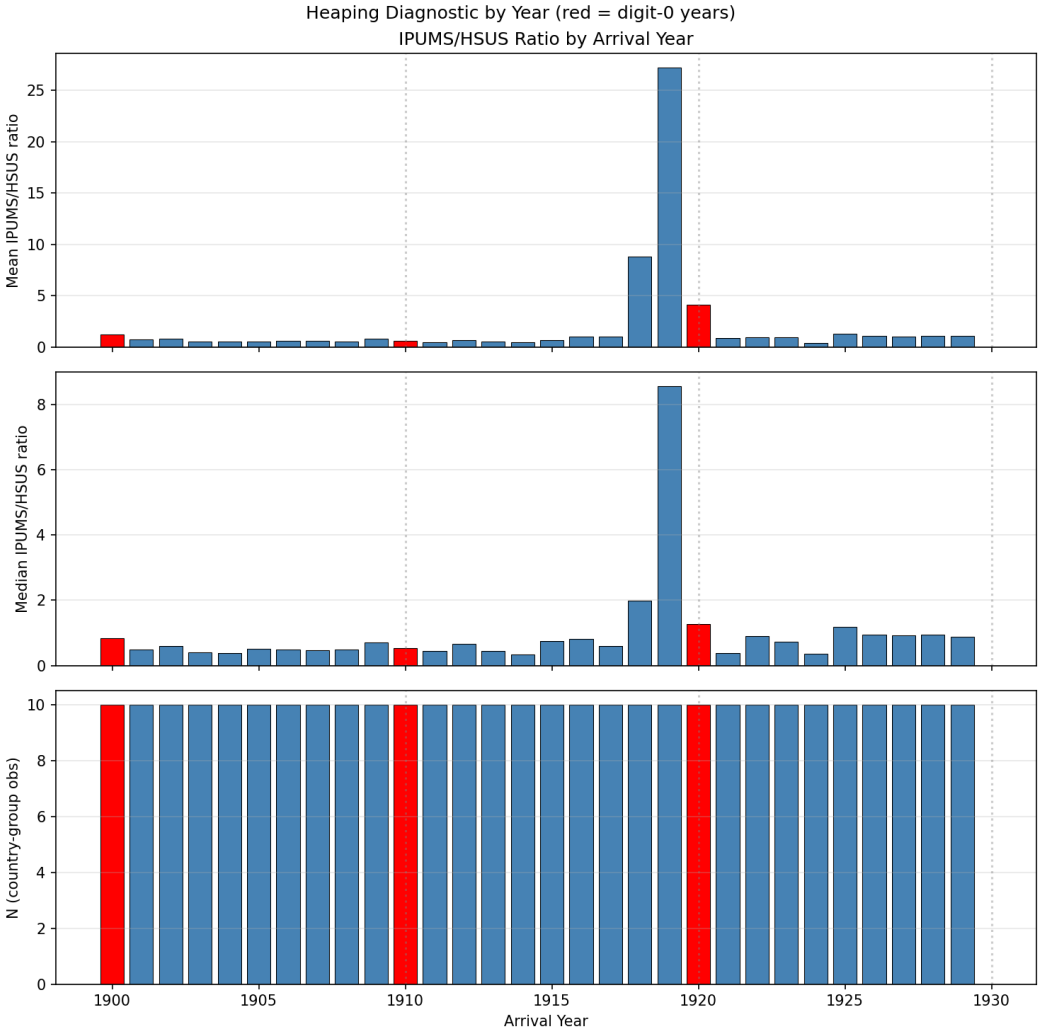


Figure 5: IPUMS/HSUS ratio by arrival year (red bars = digit-0 years). Top: mean ratio. Middle: median ratio. Bottom: N (country-group observations per year). The extreme mean ratios in 1918–1919 are driven by tiny HSUS denominators during WWI.

years is much more moderate (∼1–5), indicating the extremes are driven by a few country groups.

2. **Digit-0 years show elevated mean ratio despite maximum attrition.** Under the corrected assignment rule, all three digit-0 years (1900, 1910, 1920) are assigned to the *next* Census—10 years later—giving them maximum attrition. Despite this, the mean ratio is 2.01 (median 0.83). The elevated mean (above 1.0) is striking: attrition alone should push the ratio well below 1.0, yet heaping at round numbers inflates IPUMS counts enough to compensate. The median below 1.0 indicates that the inflation is concentrated in a few country groups, while most show the expected attrition-driven deficit.

   - **Attrition timing confound.** Under the corrected rule (arrival years $< 1910 \rightarrow$ 1910 Census; 1910–1919 $\rightarrow$ 1920 Census; 1920–1929 $\rightarrow$ 1930 Census), years 1900, 1910, and 1920 are each exactly 10 years from their assigned Census (maximum attrition). Years ending in 9 (1909, 1919, 1929) are only 1 year from their Census (minimum attrition). This systematic confound makes digit-level comparisons difficult to interpret without controlling for years-since-census.
   - **HSUS denominator variation.** Digit-0 years include peak immigration years (1900, 1910) with very large HSUS counts, which partially offsets the heaping inflation.

Table **??** reports the digit-level summary from the previous version, retained for comparison with the year-level figure.

Regression control. To disentangle heaping from the attrition timing confound, we regress ln(IPUMS/HSUS) on digit dummies (reference: digit 1) and years-since-census. Under the corrected assignment rule, the digit-0 coefficient is +0.83 but not statistically significant, indicating that heaping at digit 0 does not significantly inflate the IPUMS/HSUS ratio beyond what attrition timing explains. Note that the corrected rule changes digit-0 interpretation: years 1910 and 1920 are now 10 years from their assigned Census (high attrition), which explains the elevated mean ratio for digit 0.

3.7 Composition Preservation Test

The critical question for **endogenous variable measurement** is whether country-group composition is preserved despite heaping. The endogenous variable measures the share of immigrants in a city in a given year. If heaping affects all nationalities equally (i.e., Germans,

Table 6: IPUMS/HSUS Ratio by Last Digit of Arrival Year

Last Digit	Mean Ratio	Median Ratio	N
0	2.01	0.83	30
1	0.67	0.47	30
2	0.80	0.74	30
3	0.68	0.62	30
4	0.44	0.39	30
5	0.85	0.73	30
6	0.91	0.80	30
7	0.88	0.72	30
8	3.35	0.86	30
9	9.34	1.16	30

Ratio = IPUMS survivor count / HSUS admission count for each country-group × year cell. Digit-0 has $N = 30$ because it includes 3 years (1900, 1910, 1920) × 10 country groups (year 1930 is excluded under the corrected assignment rule).

Italians, and Russians all round at the same rate), then city-level immigrant shares are preserved even when year-level totals are noisy.

For each arrival year, we compute country-group shares from both IPUMS and HSUS, then correlate the two share vectors within each year.

The mean within-year correlation of **0.808** indicates that IPUMS reasonably preserves the country-group composition of immigration flows. The minimum correlation of 0.37 in 1918 is explainable: WWI produced very low total immigration, so small absolute errors create large share discrepancies. The fiscal year/calendar year mismatch (HSUS fiscal year ends June 30; IPUMS YRIMMIG is calendar year) is also most consequential during 1917–1919 when flows changed rapidly.

3.8 Overlap Cohort Check

As additional evidence, we compare IPUMS counts for the same arrival year observed from two different Censuses. For example, people arriving in 1905 are countable in both the 1910 Census (5 years later) and the 1920 Census (15 years later). We would expect *fewer* people in the later Census due to mortality and return migration.

Result: The mean "attrition" rate is **−293%**—the later Census reports approximately 4× more arrivals than the earlier Census for the same arrival years.

This paradoxical finding is direct evidence of heaping. The mechanism: a person who

Table 7: Composition Preservation: IPUMS vs HSUS Country-Group Shares

Statistic	Value
Mean within-year Pearson r	0.803
Min Pearson r (year)	0.370 (1918)
Max Pearson r (year)	0.966 (1912)
Number of years	30
Country groups per year	10

For each year 1900–1929, we compute the share of each of 10 country groups in total European immigration, separately from IPUMS and HSUS. The Pearson r correlates these 10-element share vectors within each year. Year 1930 excluded under corrected assignment rule.

actually arrived in 1912 may report "1910" in the 1920 Census (rounding to a prominent number). This person was *not in the US during the 1910 Census* (they hadn't arrived yet), so they don't appear in the 1910 count for arrival year 1910. But they do inflate the 1920 Census count. Additionally, longer recall distance (asking about an event 10–20 years ago vs. 0–9 years ago) increases rounding.

3.9 YRIMMIG Missing Values

YRIMMIG = 0 (missing/not reported) occurs in **0.0%** of European foreign-born records across all three census years (1910, 1920, 1930). The variable is fully populated for our sample.

3.10 Assessment: YRIMMIG Reliability

1. **Heaping exists** and is confirmed by the overlap cohort check (−293% attrition).

2. **Heaping does not affect the instrument.** The instrument is constructed from 1900 settlement shares (which do not use YRIMMIG) interacted with HSUS national flows (which do not use YRIMMIG). YRIMMIG is used only for the endogenous variable.

3. **Heaping in the endogenous variable is absorbed by IV.** Measurement error in the endogenous variable (ImmShare_{ct}) is classical—it attenuates OLS but does not bias 2SLS, which is precisely why we use instrumental variables.

4. **Country-group composition is preserved** (mean $r = 0.80$), suggesting that heaping is approximately nationality-neutral. This means the composition of immigrant arrivals at the city level is not systematically distorted, reducing the concern of non-classical measurement error.

5. **Country-group rankings agree** between IPUMS and HSUS (mean Spearman $\rho = 0.75$), further confirming that the relative magnitude of flows from different source countries is preserved.

6. **Caveat:** Ireland shows notably weak correlation ($r = 0.34$) even after correcting a crosswalk bug (BPL 421 = Northern Ireland was misclassified). The residual weakness reflects high Irish return migration rates and BPL ambiguity ("United Kingdom, ns" codes may capture some Irish immigrants). This warrants monitoring but does not invalidate the instrument, which uses HSUS flows (not IPUMS) for the national component.

4 Methodology

All analyses use a single Python script (`codes/build/13_investigate_city_yrimmig.py`) that processes the IPUMS data in two passes:

1. **Pass 1:** 1900 complete-count Census (`usa_00107.csv.gz`, 76 million records). Reads `CITY`, `METAREA`, `BPL`, `NATIVITY`. Accumulates counts for Q1 analysis using dictionary-based accumulation for memory efficiency.

2. **Pass 2:** 1910–1930 complete-count Censuses (`usa_00108.csv.gz`, 320 million records). Reads `CITY`, `METAREA`, `BPL`, `NATIVITY`, `YRIMMIG`. Accumulates counts for both Q1 (city/metarea) and Q2 (YRIMMIG by country group and census year).

Processing uses 2-million-row chunks with dictionary accumulators, keeping memory under 100 MB. Total runtime: approximately 15 minutes. Country-group mapping follows the same BPL-to-group dictionary used in the pipeline (`08b_process_ipums_settlement_shares.py`).

A `--figures-only` mode regenerates all figures from existing CSV outputs without reprocessing the IPUMS microdata.

5 Limitations and Next Steps

1. **Geographic mismatch.** The outcome variable (Lyons house prices) is at the city-proper level, while the explanatory variables use `METAREA`. This mismatch is mitigated

by the high settlement share correlation ($r = 0.978$) but should be tested via a robustness check using `CITY`-based shares for the 26 identifiable cities.

2. **HSUS fiscal year mismatch.** HSUS immigration data uses a fiscal year ending June 30, while YRIMMIG is calendar year. This creates ∼6-month misalignment, most consequential during years with sharp policy changes (1914 WWI onset, 1921 Quota Act).

3. **Ireland/Great Britain classification.** The weak Ireland correlation ($r = 0.46$) and evidence of elevated Great Britain IPUMS shares relative to HSUS suggest possible BPL coding issues. This should be investigated before treating Irish settlement shares as precisely measured.

4. **Non-classical heaping risk.** While the composition preservation test ($r = 0.80$) is reassuring, it is conducted at the *national* level. Heaping could in principle be more severe in certain cities or for certain subpopulations. City-level heaping diagnostics would be more direct.

5. **Robustness.** Re-estimating the 2SLS with `CITY`-based settlement shares (for 26 cities) would provide a direct robustness check on the main results.

6 Human–AI Collaboration

Human researcher (Guohui Jiang):

- Identified both data quality concerns via PDF annotations on Progress Report #6
- Reviewed v1 of this report and identified six issues via further annotations:
  - Missing conceptual evaluation of CITY vs METAREA (not just empirical)
  - Decision rationale should center on Lyons compatibility
  - Need for individual country-group flow figures and ranking comparison
  - Need for year-level heaping figures (not just digit-level)
  - Critical correction: YRIMMIG affects endogenous variable, not instrument
- Reviewed v2 and identified incorrect Census assignment rule: boundary years (1910, 1920) must go to the *next* Census because Census Day falls partway through the year

- Reviewed v3 and identified: (a) weak METAREA arguments to remove, (b) need to investigate Ireland's weak correlation and audit the BPL crosswalk, (c) need to list the crosswalk verbatim in the report
- Directed the investigation workflow: plan → agent review → code → agent review → run → report → annotation → revision (4 cycles)

Claude Code (AI assistant):

- Designed the single-pass data processing strategy for 396 million total records
- Wrote `13_investigate_city_yrimmig.py` with dictionary-based accumulation
- Generated all tables, figures (including 10 individual country-group flow figures, ranking comparison, and year-level heaping diagnostic), and all four versions of this report
- Corrected conceptual error about instrument vs endogenous variable across 6 repo files (v2)
- Corrected Census assignment rule and re-ran analysis with corrected boundaries (v3)
- Found and fixed crosswalk bugs: BPL 421 (Northern Ireland) misclassified as "Belgium," BPL 435 (Italy, n.s.) unmapped. Added crosswalk table to report (v4)

References

Tabellini, M. (2020). "Gifts of the Immigrants, Woes of the Natives: Lessons from the Age of Mass Migration." *Review of Economic Studies*, 87(1), 454–486.